

Does the 1B algorithm transfer to 1.5B?

`iter_R4_grow_on_sampled_uniform_M5`
re-run on the new 1.5B (step-49999)

Short answer: no.

1B: +42.81% over baseline

1.5B: only +9.04% — and stage A alone is *worse* than baseline.

Same 10 train proteins · same hyperparameters · A100 40GB bf16

Setup

- ▶ Rebased `exp/27-improved-inference-algorithm` onto main → `MODELS.yaml` now lists 1.5B
(`buckets/open-athena/MarinFold/.../protein-contacts-1_5b-distance-masked-70f8f5/step-49999`).
- ▶ Architecture: **24 hidden layers** (vs 1B's 16), hidden 2048, **8 KV heads / 32 query heads** (GQA), intermediate 8192. $\sim 1.5\times$ the parameters of 1B.
- ▶ Checkpoint name `step-49999` suggests **undertrained** relative to the 1B production checkpoint.
- ▶ Algorithm and knobs held **exactly identical to the 1B headline**:
`sampled_uniform_M5_union` prior + `iter_R4_grow` on top,
`range_strategy=uniform`, `kc=[0.5, 1, 1.5, 2.5]`, `min_contact_prob=0.1`,
standard expected-distance readout, no sharpening.
- ▶ Same 10-protein train set, output to `outputs_1.5b/` so 1B snapshots are preserved.

Headline numbers

run	mean LDDT	median	wall (s)	lift
1B baseline (naive)	0.2496	0.2273	1387	—
1B combined	0.3564	0.3560	3155	+42.81%
1.5B baseline (naive)	0.2627	0.2315	1866	—
1.5B stage A only	0.2038	0.2126	3472	-22.42%
1.5B combined	0.2864	0.2545	4230	+9.04%

stage A = `sampled_uniform_M5_union`; combined = `iter_R4_grow_on_sampled_uniform_M5`.

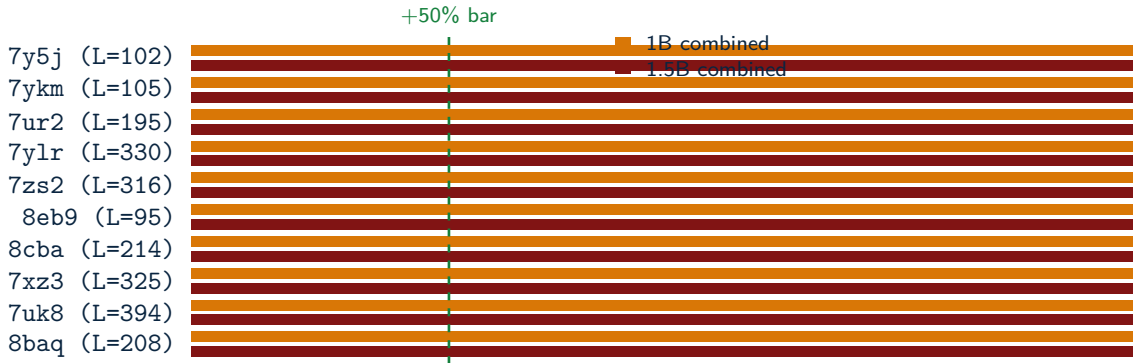
- ▶ 1.5B *baseline* is only modestly better than 1B (+5%).
- ▶ 1.5B with the algorithm gives **0.2864** — worse in absolute terms than the 1B combined result (0.3564) and the lift over its own baseline drops from **+42.81%** → **+9.04%**.
- ▶ **Stage A alone (`sampled_uniform_M5_union`) is below the 1.5B baseline** — sampled-uniform contacts hurt the 1.5B model.

Per-protein lift (combined vs same-model baseline)

stem	L	1B base	1B combined	1B lift	1.5B combined	1.5B lift
8eb9	95	0.1510	0.3071	+128.8%	0.2665	+78.1%
7y5j	102	0.4485	0.5136	+14.6%	0.5645	+6.0%
7ykm	105	0.3317	0.4639	+39.9%	0.4010	+28.4%
7ur2	195	0.2633	0.3665	+39.2%	0.3090	+19.9%
8baq	208	0.1872	0.2567	+37.1%	0.2425	+30.8%
8cba	214	0.2045	0.3028	+48.1%	0.2772	+35.0%
7zs2	316	0.2500	0.3778	+51.1%	0.2086	-25.7%
7xz3	325	0.1913	0.2727	+42.6%	0.2146	+6.7%
7ylr	330	0.2673	0.3991	+49.3%	0.2140	-30.3%
7uk8	394	0.2010	0.2657	+32.2%	0.1661	-14.6%

Pattern: the 3 regressions on 1.5B are the 3 longest proteins (L=316, 330, 394). Small / mid proteins still benefit but with smaller gains than on 1B. The mean is dragged down by the long-protein failures.

Visual: 1B combined (orange) vs 1.5B combined (red)



Long proteins on 1.5B (7ylr, 7zs2, 7uk8 — bottom) drop dramatically. 7y5j and 8cba are the only proteins where 1.5B beats 1B.

Where it breaks: stage A makes things worse

	1B	1.5B	delta
baseline (naive readout)	0.2496	0.2627	+0.013
stage A only (sampled_uniform_M5_union)	0.3142	0.2038	− 0.111
stage A lift vs baseline	+25.9%	−22.4%	
combined (stage A → stage B)	0.3564	0.2864	−0.070
combined lift vs baseline	+42.8%	+9.0%	

- ▶ Stage A delivers a +0.065 boost on 1B but a **−0.059 regression** on 1.5B — a sign-flip of ~ 0.12 absolute LDDT.
- ▶ Stage B (iterative growing K) still helps relative to the (poor) stage-A prior, but it can't recover the gap.
- ▶ The sampled-uniform contact prior, which was the algorithm's main novelty on 1B, is actively harmful on 1.5B.

Why?

- ▶ **Hyperparameters tuned on 1B don't generalize.** `range_strategy=uniform` was specifically the fix for 1B's ~99% medium-range prior. 1.5B may have a different range-token prior — if so, forcing uniform actively hurts. The entropy probe was run on 1B; we haven't re-run it on 1.5B.
- ▶ **1.5B is undertrained.** Checkpoint name is `step-49999`; the 1B is a production checkpoint that's seen far more training. The 1.5B may not yet have learned the conditional-distance distribution the algorithm relies on.
- ▶ **Long-protein failure mode.** 1.5B regresses on the 3 longest proteins ($L \geq 316$). Possible cause: under GQA + more layers, the model's response to long contact-prefix prompts is different, and our K schedule was tuned for the 1B's attention pattern. Long proteins get the largest seed prefix ($K=2.5L$ = hundreds of statements) — most impacted by any context-handling mismatch.
- ▶ **The 1.5B baseline itself is only modestly better than 1B (+5%).** If the bigger model is just slightly stronger naively, the algorithm's lift has less to push against — but that doesn't explain the sign-flip on stage A.

Takeaways and next steps

- ▶ **The 1B algorithm is not model-portable as-is.**

`iter_R4_grow_on_sampled_uniform_M5` gives +42.81% on 1B and only +9.04% on 1.5B (step-49999) with identical knobs.

- ▶ **Stage A (sampled-uniform contacts) is the broken piece.** It's a +25.9% lift on 1B and a -22.4% regression on 1.5B. The "force uniform range" trick was the 1B-specific fix to that model's specific range-token bias.
- ▶ **Stage B (iterative growing K) is more portable.** It still adds lift even from a bad stage-A prior. Iteration probably wants different K-schedule numbers per model.
- ▶ **Per-model tuning would help.** Concretely:
 - ▶ Re-run `probe_range_entropy.py` on 1.5B to measure its range-token prior + conditional position entropy.
 - ▶ Try `range_strategy=model` on 1.5B in case its natural prior is already balanced.
 - ▶ Try smaller K (e.g. K=0.5L cap) on long proteins where prefix-handling appears to misbehave.

Reproducer

```
# 1.5B: stage A
uv run python run_sampled_contacts.py --dtype bfloat16 --n-gpus 1 \
    --out outputs_1.5b --scores-out data/sampled_contacts_scores_1.5B.csv \
    --model 1.5B --algorithm sampled_uniform_M5_union_T0.7_1.5B \
    --n-rollouts 5 --temperature 0.7 --top-p 0.9 --max-sample-tokens 900 \
    --range-strategy uniform --aggregation union
uv run python snapshot_distograms.py --out outputs_1.5b \
    --to histogram_sampled_uniform_M5_union_1.5B.npz

# 1.5B: stage B
uv run python run_iterative.py --dtype bfloat16 --n-gpus 1 \
    --out outputs_1.5b --scores-out data/iterative_scores_1.5B.csv \
    --model 1.5B --algorithm iter_R4_grow_on_sampled_uniform_M5_1.5B \
    --n-rounds 4 \
    --k-contacts-per-L-per-round 0.5 1.0 1.5 2.5 \
    --k-distances-per-L-per-round 0.0 0.0 0.0 0.0 \
    --min-contact-prob 0.1 \
    --prior-name histogram_sampled_uniform_M5_union_1.5B.npz

# Results:
# 1.5B sampled_uniform_M5_union_T0.7_1.5B:    0.2038  (-22.4% vs 1.5B baseline)
# 1.5B iter_R4_grow_on_sampled_uniform_M5_1.5B: 0.2864  (+9.04% vs 1.5B baseline)
```